

Empowering one-vs-one decomposition with ensemble learning for multi-class imbalanced data



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ABSTRACT

Multi-class imbalance classification problems occur in many real-world applications, which suffer from the quite different distribution of classes. Decomposition strategies are well-known techniques to address the classification problems involving multiple classes. Among them binary approaches using one-vs-one and one-vs-all has gained a significant attention from the research community. They allow to divide multi-class problems into several easier-to-solve two-class sub-problems. In this study we develop an exhaustive empirical analysis to explore the possibility of empowering the one-vs-one scheme for multi-class imbalance classification problems with applying binary ensemble learning approaches. We examine several state-of-the-art ensemble learning methods proposed for addressing the imbalance problems to solve the pairwise tasks derived from the multi-class data set. Then the aggregation strategy is employed to combine the binary ensemble outputs to reconstruct the original multi-class task. We present a detailed experimental study of the proposed approach, supported by the statistical analysis. The results indicate the high effectiveness of ensemble learning with one-vs-one scheme in dealing with the multi-class imbalance classification problems.

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1. Introduction

In machine learning and data mining, while one or more classes are underrepresented in the data set, it is called as class imbalance classification. Many real-world classification tasks suffer from the class imbalance problem, which is considered as one of the important challenges for the data mining community [18]. The main difficulty of these problems is that the skewed distribution makes conventional classification algorithms less effective, since standard learning algorithms consider a balanced training data set, which result in making it harder to predict minority class examples [50].

In recent years, many efforts have been focused on the binary class imbalance problems [31,41], which only contain two classes. However, multi-class imbalance classification, is widely applied in

many areas, such as text categorization [47], human activity recognition [1] and medical diagnosis [35]. Unfortunately, it may be invalid to directly apply the solutions proposed for the two-class problems to the multi-class imbalance problems, and some algorithms cannot be used to solve the multi-class imbalance problems directly [18].

Fortunately, in the research community, decomposition strategies turn up to deal with multi-class classification problem. In this solution framework, the multi-class classification problems are transformed into binary class sub-problems, which are much easier to discriminate [53,61]. Such well-known approaches are the one versus one (OVO) [23,33] and one versus all (OVA) [7]. As OVA introduces an artificial class imbalance (e.g., for 10 class problem with roughly equally represented classes, the binary sub-problem will have an imbalance ratio 1:9), it is not advisable to use it for handling problems with initially skewed distributions [46].

In this paper, we focus on multi-class imbalance classification problems and develop a complete empirical study to explore the effectiveness of ensemble learning methods [62] in the multi-class

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imbalanced datasets with OVO scheme, where binary-class classifiers are trained from the subset containing each pair of classes by ensemble learning approaches based on data preprocessing [22]. Our initial works in this domain showed that empowering binary decomposition with pairwise ensemble learning can significantly improve mining imbalanced multi-class problems [34].

Regarding ensemble learning methods, six state-of-the-art approaches are selected to carry out the experiment: UnderBagging [3], SMOTEBagging [15,56], RUSBoost [49], SMOTEBoost [10], SMOTE+AdaBoost [40], EasyEnsemble [40]. Additionally, to show the efficiency of ensemble learning with OVO scheme for addressing the multi-class imbalance problems, the original data preprocessing strategies, including random under-sampling (RUS) [4], random over-sampling (ROS) [4] and synthetic minority oversampling technique (SMOTE) [8], are also implemented in the OVO scheme for our comparative analysis.

Finally, we carry out a thorough experimental study that supports the effectiveness of our methodology. Concretely, 20 multi-class imbalanced data sets are selected from the UCI repository in our experiment. The average accuracy rate [20] is used as the performance measures in this study. In order to analyze the results obtained from the different solutions, statistical analysis suggested in [28] is given to support the significance of the results.

The main contributions of this paper with respect to previous studies are as following:

- We propose to enhance the OVO scheme for multi-class imbalanced data by using ensemble techniques for each sub-problem.
- We show, how to extend the area of applicability of binary imbalanced ensemble classifiers to handling far more challenging multi-class imbalanced scenarios.
- We develop a complete experimental study of comparison of the state-of-the-art ensemble learning techniques with conventional resampling methods with OVO strategy and state-of-the-art solutions for multi-class imbalance problems.
- In order to obtain the impacts of the base classifier used in our scenario, we choose three different algorithms, including Classification and Regression tree (CART) [6], Back Propagation Neural Network (BPNN) [17] and Support Vector Machine (SVM) [54].

The rest of this paper is organized as follows. The background of this study is introduced in Section 2, including multi-class imbalance classification problems and decomposition strategies. Next, in Section 3 we present the framework of our methodology of ensemble learning with OVO scheme for dealing with multi-class imbalance classification problems. In Section 4, the experimental framework is given, including the data sets, the base classifiers and the relative parameters setting, the performance measures and the statistical tests. The complete empirical study is presented in Section 5. Lessons learned from the paper are given in Section 6, while conclusions and potential directions for future works are to be found in the final Section.

2. Background

In this section, we first introduce the problem of multi-class imbalance classification. Then, we present the solutions for addressing the imbalance problems. Finally, we describe the decomposition strategy for dealing with multi-class classification problems.

2.1. Multi-class imbalanced data analysis

Multi-class imbalanced data sets, where there are much more instances of some classes (referred to as the majority classes) than

others (referred to as the minority classes), is one of the most challenging problems with data quality that always reduces classification performance in machine learning and data mining [57]. The minority classes are usually the most important concepts to recognize, since they represent the rare cases [59]. Additionally, it is expensive or hard to select these examples [58].

However, standard classification algorithms are designed with the premise of a balanced training set [42]. With such a precondition, it is much more difficult for the classical classification algorithms to deal with class imbalance problems, especially for identifying the minority class instance [9]. Additionally, most of the methods however are specific to address the binary class imbalance problems. Obviously, multi-class imbalance problems are far more complex, since these issues are involved with large number of classes and the relationships among the classes are complicated. Furthermore, it is hard to distinguish between minority classes and noise examples and the minority classes can be ignored by the classifier as the noise examples.

2.2. Solutions for imbalanced classification problems

To overcome the dilemma of skewed class distribution, a large amount of techniques have been developed to deal with such problem. These proposals can be roughly categorized into four groups:

- Data level: the origin of the problem is the class distribution in the data sets, therefore, it is natural to consider of rebalancing by sampling the data space to reduce the impact of class imbalance, known as an external approach. One of the advantages of such solution is independent from the classifier used, so they are also considered as pre-sampling method [27,51].
- Algorithm level: these solutions try to adopt appropriate decision threshold to reinforce the learning towards the minority class instances. The proposed algorithms that take the class imbalance into consideration belong to such techniques. They are defined as internal approaches in some papers [11,52], since the effect depends on the problems and the classifier [13]. One of the most well-known solutions is the direct modification of the learning procedure for a selected algorithm [45].
- Cost-sensitive level: these approaches consider higher costs for misclassifying the minority classes with respect to the majority classes, that is, misclassification of minority class is much more expensive [44]. The learning process turns to minimize the cost errors instead of maximization of accuracy rate [63].
- Ensemble level: these solutions combine the efficient ensemble learning solutions [62] with one of the three previously mentioned strategies in order to create a balanced training sets for base classifier and at the same time introduce diversity into the pool of base learners. Special attention should be paid to recent combination of intelligent and directed data-level approaches with Bagging solution [5] or randomized oversampling [15], hybrid combination of algorithm-level methods [55] and cost-sensitive pruning for decision tree ensembles [36].

Due to the advantage of the data level solutions (as pointed out by a recent tutorial on data preprocessing [29]) we focus on such methods in this study.

RUS [51] is the basic under-sampling, which randomly removes the majority class instances to balance the class distribution. This approach is efficient for dealing with class imbalance problems, since most of the majority class instances are redundant. Additionally, RUS makes the training process become much faster, since the training set contains less instances than original data set. However, some potential useful information contained in the majority class instance may be neglected, since RUS randomly generates the subset without considering the relationship among the instances.

ROS [27] is another basic resampling approach, which randomly duplicates the minority class instances to make the training set balanced. With such consideration, the size of minority class examples tends to match the size of the other classes. However, it might lead to two important defects. One of the problems is overfitting, since ROS typically replicates examples of the minority class so that the interface between minority class and majority class is too prone to the former. Another problem is that ROS increases the training time. Suppose there is an imbalance data set with 20000 majority class examples and 100 minority class examples, to generate an equal size of minority class and majority class, the ROS will create a training data set with 40000 examples. Obviously, it must take much more time if ROS is used as resampling technique.

SMOTE is an intelligent oversampling approach proposed by Chawla et al. [8]. Unlike ROS duplicates the minority examples, SMOTE produces synthetic minority class examples by k nearest neighbors, augmented with randomized interpolation. However, the noise might incorporate in the synthetic minority class examples.

In the above reviewed methods, sampling process is independently carried out before the training process. In these approaches, the training process is unchanged and the classifiers are trained by the balanced data sets produced in the resampling process. To overcome the drawbacks of classical resampling approaches mentioned above, some ensemble learning methods [37] are proposed to devote to combine the resampling process and training process. These techniques developed for dealing with class imbalance problems follow the architecture of bagging or boosting.

In the framework of bagging, the base classifier is trained by using the resampling examples. That is, the diversity is obtained through resampling minority class instances, since bootstrapped instances of the original data set are randomly duplicated. Many proposals are introduced in the research community, such as OverBagging, UnderBagging, and UnderOverBagging [3].

Regarding to the scenario of boosting, the resampling techniques are embed into bootstrap. In such a manner, the resampling methods are introduced in each iteration to train a classifier toward the minority class. In this family, SMOTEBoost [10] and RUSBoost [49] are the representative proposals. Besides these techniques, some algorithms combine both bagging and boosting to obtain an ensemble of ensembles [40].

However, when we refer to multi-class imbalance classification problems, the conventional solutions may not be feasible or perform a lower achievement. There are still only few approaches for handling this task. Static-SMOTE [19] applied resampling procedure in M steps, where M is the number of classes. In each iteration, the resampling procedure selects the minimum size class, and duplicates the number of instances of the class in the original data-set. An ensemble learning algorithm for multi-majority and multi-minority cases was proposed in [57]. Authors combine AdaBoost with negative correlation learning, where starting weights of examples are calculated in inverse proportion to the number of objects in this class. A combination of binary decomposition and pre-processing methods was proposed as an efficient solution when the number of classes is high [18].

More recent studies on this problem propose to combine pairwise modified SVMs with boundary shift asymmetric regularization costs using one-vs-all technique [14]. Additionally, two hybrid ensemble techniques with embedded feature selection were proposed to deal with the problem of skewed distributions among multiple classes [30,38]. The most recent work study in this domain reports the high importance of considering the individual types of minority classes examples and their learning difficulty when performing oversampling for multi-class imbalanced data and proposes a data-driven universal strategy that can be embedded in any data-level multi-class solution [48].

As one can see, there are many efficient ensemble techniques dedicated to binary problems. However, it is not straightforward to extend them for scenarios with higher number of classes. Therefore, the ensemble learning methods are considered in this study to learn the pattern from the data set derived from paired classes, and then the binary ensembles are combined by aggregation strategy to be a final multi-class classifier.

2.3. One-vs-one scheme for multi-class classification problems

In the OVO decomposition scheme, a m -class problem is divided into $m(m-1)/2$ binary subproblems. Each problem is faced by independent base classifiers, which are responsible for distinguishing the instances from the different pairs of classes. With such appropriate consideration, the much more complex multi-class problem is translated into the simpler binary class subproblems, which is expected to receive better results or address multi-class problems with binary classification techniques. An example of binarization technique for decomposing the multi-class problem is shown in Fig. 1.

In order to predict a new pattern, there are two phases required to complete the task. The first phase is to learn the classifiers that are trained by the original instances with pairwise classes, that is, the task of each classifier is to distinguish a pair of classes $\{C_i, C_j\}$. In the integration phase, a confidence degree $r_{ij} \in [0, 1]$ in favor of C_i is given by a classifier to discriminate the class i from class j . The confidence in favor of j -th class is computed as $r_{ji} = 1 - r_{ij}$, if the classifier considers the output class as the class with the largest confidence value. To be clearer, all the confidence degree can be represented by a score matrix R :

$$R_{m,n} = \begin{pmatrix} - & r_{1,2} & \cdots & r_{1,m} \\ r_{2,1} & - & \cdots & a_{m,n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m,1} & r_{m,2} & \cdots & - \end{pmatrix}. \quad (1)$$

There are a number of aggregations proposed in the literature [26,39] to infer the final class. The voting strategy (VOTE) is the simplest but powerful aggregation, therefore, it is considered as the aggregation approach in this study. VOTE, which is also called binary voting and Max-Wins rule [21], considers a vote for the predicted class by the binary classifier. Votes received by each class are counted and the final class obtaining the largest number of votes is predicted as follows:

$$\text{Class} = \arg \max_{i=1,\dots,m} \sum_{1 \leq j \neq i \leq m} s_{ij}, \quad (2)$$

where

$$s_{ij} = \begin{cases} 1 & r_{ij} > r_{ji} \\ 0 & \text{otherwise} \end{cases}. \quad (3)$$

3. Combining OVO with ensemble learning for multi-class imbalanced data

As mentioned above, on the one hand, ensemble learning proposed for imbalanced datasets may be not effective or even impossible for dealing with multi-class problems. On the other hand, decomposition strategy is effective direction to handle the multi-class classification problems. Therefore, it is interesting for us to consider their respective advantages in the scenario of multi-class imbalanced datasets. In this section, we firstly describe our methodology of employing the ensemble learning approaches in the OVO scheme. Then, we present the well-known ensemble learning methods employed in this study.

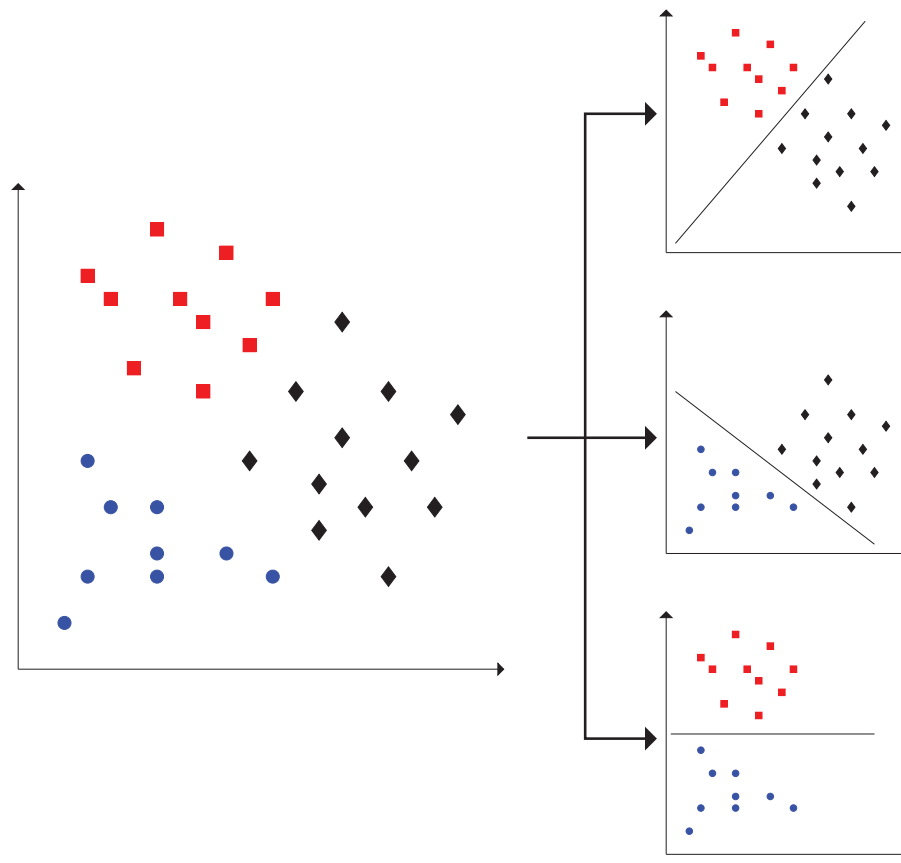


Fig. 1. An example of one-versus-one (OVO) decomposition of a three-class problem into three two-class problems.

3.1. Solving multi-class imbalanced problems with binary decomposition

Multi-class imbalance classification problems are extremely complex tasks, which suffer from more than two classes and class imbalance distribution. According to the suggestion of decomposition strategy, which aims at solving the complex problem by decomposing it into series of much easier subproblems, we propose to explore the effectiveness of combination of ensemble learning and OVO scheme to address the multi-class imbalance classification problems.

Firstly, the multi-class imbalance classification problems are decomposed into a series of pairwise datasets. One should note that there are some pairwise classes after decomposition will have roughly equal size. For these cases, the subproblems are handled as normal binary classification problems by the normal ensemble learning algorithms. Then, the ensemble learning methods based on the resampling techniques devote to address the binary class imbalance datasets.

In this work, we focus only on OVO approach as recent studies on decomposition techniques clearly proved its superiority over OVA methods [24,25]. Additionally, studies on combination of resampling strategies with single classifiers for multi-class imbalanced data showed clearly that OVA displays inferior performance [18]. Finally, OVA introduces an artificial class imbalance which may further damage the learning process in scenarios with multiple skewed distributions.

An example of the proposed approach is shown in Fig. 2.

There are two strategies employed in the ensemble learning process.

As shown in Fig. 2 (left), one of the ensemble learning algorithms combines the single classifiers which are derived from

the balanced datasets by reducing the size of the majority class. Concretely, the multi-class classification problem is split into several binary class subproblems. Then, the ensemble learning method combining under-sampling strategy is employed to learn a binary classifier for each subproblem. Finally, the aggregation strategy is considered to obtain the final output class.

As described in Fig. 2 (right), oversampling approach is adopted to increase the number of the minority class to balance the class distribution. Firstly, the binary class subproblems are derived from the original multi-class classification problem. Secondly, ensemble learning method considering SMOTE to create the synthetic minority class to balance the distribution for the training process is used to achieve a binary classifier for each pairwise class data set. Finally, once each binary classifier derived from the ensemble learning, aggregation strategy will be employed to provide the final output from the score matrix.

Let us now discuss ensemble learning algorithms used in this study.

3.2. Ensemble learning algorithms for binary imbalanced data

As described above, ensemble learning is applied in our study to classify the binary classes which are derived from the multi-class data sets. Therefore, several well-known ensemble learning algorithms proposed for imbalance problems were selected to train the binary classifiers. Specifically, the ensemble learning algorithms employed in this study are as follows:

- UnderBagging (UBA). RUS is applied to under-sampling the majority class in each bag of the ensemble. In this way, we can obtain the balanced data set for each classifier of the bag and the diversity for each bag is produced with different majority

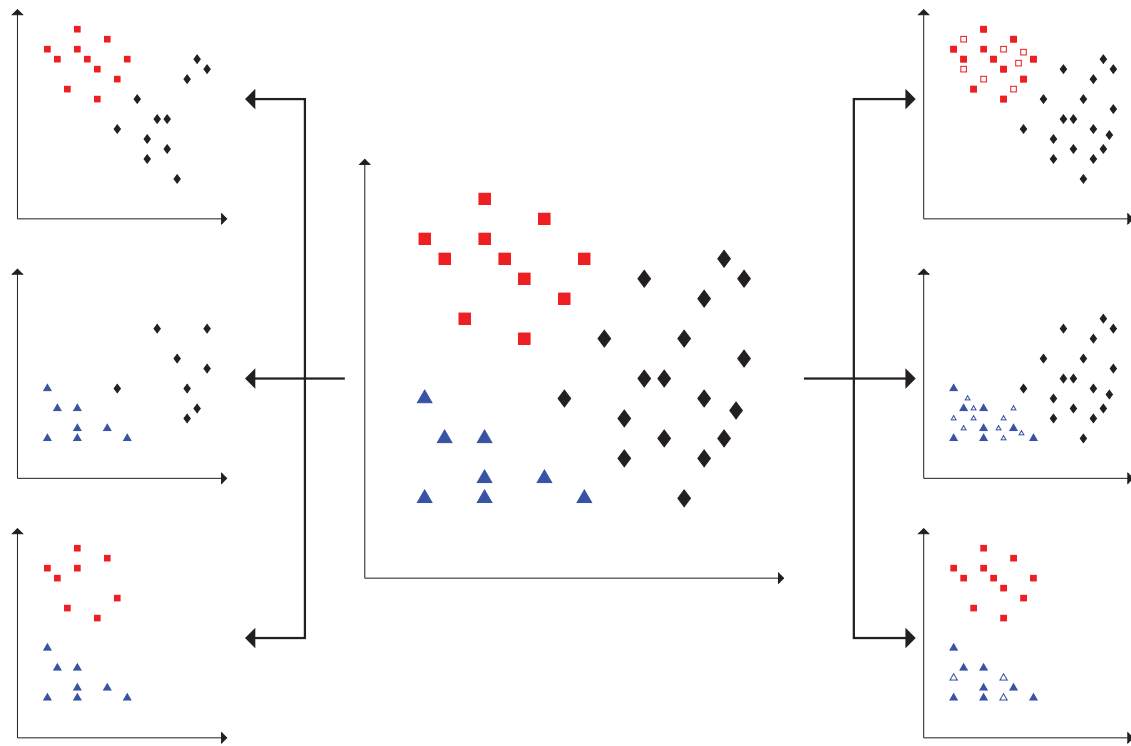


Fig. 2. An example of pre-processing methods applied over a decomposed multi-class imbalanced dataset. (Left) Undersampling applied to each subproblem. (Right) Oversampling applied to each subproblem.

class instances and the same minority class instances. Additionally, the strategy of resampling with replacement is adopted in order to increase the diversity among the bags.

- **SMOTEBagging (SBA).** SMOTE algorithm is employed to insert artificial minority class in each bag. This way aims at balancing the distribution of the data set by increasing the number of minority class instances. The diversity of each bag of the ensemble is obtained with mutative synthetic samples.
- **RUSBoost (RBO).** This strategy applies RUS of minority class to balance the distribution of the data set before ensemble learning. The AdaBoost is employed to construct a classifier from the balanced data set. One should note that this method handle the imbalance problem before the ensemble learning process.
- **SMOTEBoost (SBO).** SMOTE algorithm is employed to balance the distribution of the data set in each iteration of AdaBoost. Specifically, a training set is obtained by resampled from weighted samples. Then, synthetic samples are generated to increasing the number of minority class. Therefore, the distribution of the training set for each single classifier in AdaBoost is balanced.
- **SMOTE+AdaBoost (SMB).** Considering to this method, we apply SMOTE algorithm to generate synthetic majority class to balance the data set before AdaBoost. That is, the SMOTE is independent from AdaBoost, which is different from SBO employing SMOTE algorithm during Boosting progress.
- **EasyEnsemble (Easy).** Under-sampling is simple but effective for handling imbalance problem, since the training data set is much more balanced and the training process is much faster. However, many majority class examples are ignored, which leads to loss potentially useful information. In order to overcome the drawback, this method constructs classifier ensembles from all of the minority class and a subset of the majority class. Then, the final ensemble combines the outputs of classi-

fiers which are built by using AdaBoost algorithm. Therefore, it can be considered as an ensemble of ensembles.

4. Experimental framework

In this section, we present the set-up of the experimental framework used in the experiment of Section 5. The data sets chosen to test the algorithms are described in Section 4.1, while single and ensemble classifiers with their parameters are described in Section 4.2. The measure to evaluate the performance of the approaches in this study are presented in Section 4.3. Final Section describes the statistical test applied to compare the results obtained from experiments.

4.1. Data sets

In this study, twenty data sets from the UCI repository were selected to test the methodology. The properties of the data sets were showed in Table 1. For each data set, it includes the number of examples (#Ex.), the number of attributes (#Atts.), the number of numerical (#Num) and nominal (#Nom) attributes, the number of classes (#Cl.), the distribution of class (#Dc) and the imbalance rate (IR). For the missing values instances in the data sets (Cleveland and Dermatology), we removed them before doing the partitions.

The results of the average accuracy was obtained by means of five times five-fold stratified cross-validation (SCV) [43]. That is, each data set is split into five folds and each fold contained 20% of the instances of the data set. For each fold, the algorithm is trained with the instances contained in the remaining folds (80% of the instances of the data set) and then tested by the current fold. The reason why five-fold SCV is more appropriate than a 10-fold SCV in such framework was explained in [42]. If we use smaller partitions, there would be some test sets without containing any instance from some of the classes.

Table 1

Summary description of the data sets used in the experimental study.

id	Dataset	#Ex.	#Atts.	#Num.	#Nom.	#Cl.	#Dc.	IR
Aut	Automobile	159	25	15	10	6	46/29/13/48/20/3	16.00
Bal	Balance	625	4	4	0	3	288/49/288	5.88
Car	Car	1728	6	0	6	4	384/69/1210/65	18.62
Cle	Cleveland	297	13	5	8	5	160/54/35/35/13	12.31
Con	Contraceptive	1473	9	6	3	3	629/333/511	1.89
Der	Dermatology	358	34	1	33	6	111/60/71/48/48/20	5.55
Fla	Flare	1066	11	0	11	6	147/211/239/95/43/331	7.70
Gla	Glass	214	9	9	0	6	70/76/17/29/13/9	8.44
Hay	Hayes-roth	160	4	4	0	3	65/64/31	2.10
Led	Led7digit	500	7	0	7	10	45/37/51/57/52/52/47/57/53/49	1.54
Lym	Lymphography	148	18	3	15	4	4/61/81/2	40.50
New	New-thyroid	215	5	5	0	3	150/35/30	5.00
Pag	Page-blocks	5472	10	10	0	5	4913/329/28/87/115	175.46
Sat	Satimage	6435	36	36	0	6	1533/703/1358/626/707/1508	2.45
Shu	Shuttle	2175	9	9	0	5	1706/2/6/338/123	853.00
Spl	Splice	3190	60	0	60	3	767/768/1655	2.16
Thy	Thyroid	720	21	6	15	3	17/37/666	39.18
Win	Wine	178	13	13	0	3	59/71/48	1.48
Wqr	Wine-Quality-Red	1599	11	11	0	6	10/53/681/638/199/18	68.10
Zoo	Zoo	101	16	0	16	7	41/20/5/13/4/8/10	10.25

4.2. Classification algorithms

In this section, the base classification algorithms compared in the experiment are presented. Specially, CART, BPNN and SVM are selected for the study, and they are described as follows.

CART is a tree induction technique. CART is a binary recursive partitioning methodology to generate a tree, that is, a node in the tree always split the attribute into only two parts (nodes). In CART, the Gini index is used to measure the impurity of node. The attribute and the corresponding binary split on that maximizes the reduction in impurity are chosen as the splitting point. The process will continue until reach the stopping criterion, including there is no possible split point or the maximal tree is obtained. Due to the noise or outlier in the training data set, some branches reflect abnormal information, which may lead to overfit the data. Pruning is an effective way to avoid overfitting. In CART, the cost complexity pruning algorithm which is a post-pruning method is used to prune the tree. In the approach, the cost complexity of a tree is measured by a function of error rate, which is the percentage of instances misclassified by the tree.

BPNN is a typical feed forward neural network, which has input layer, hidden layer and output layer. Back propagation learns by iteratively processing a set of training samples, comparing the networks prediction for each sample with the actual known target value. For each training sample, the weights are modified so as to minimize the mean-square error between the networks prediction and the actual target value. These modifications are made in the backwards direction through each hidden layer down to the first hidden layer.

SVM is an effective machine learning method, which is based on Vapnik–Chervonenkis structural risk minimization instead of the empirical risk. SVM maps the original input feature space into a high dimensional feature space in order to construct an optimal separating hyperplane with maximal margin. The choice of a proper kernel has a strong effect on the final quality of the obtained model [12].

These classifiers are used as base learners for six ensemble methods described in Section 3.2.

Additionally, to offer a fair comparison with other ensemble techniques dedicated to multi-class imbalanced learning we have selected two state-of-the-art methods: Near-Bayesian Support Vector Machines (NBSVM) [14] and multi-class imbalanced Boosting [57].

The first approach modifies the SVM classifier by using samples from two classes in order to achieve boundary shift and combines it with the asymmetric regularization costs. Additionally, authors propose a modification of the popular SMO algorithm to train NB-SVM. To handle multi-class scenarios a binary aggregation scenario with OVA technique is being used.

AdaBoost.NC combines the multi-class AdaBoost algorithm with negative correlation learning, where starting weights of examples are calculated in inverse proportion to the number of objects in this class. This way it is possible to capture varying relations between classes.

Detailed parameters of used methods are given in Table 2.

4.3. Performance measures

There is a large amount of measures for the performance of algorithms in the imbalance classification problems, for example, precision, sensitivity, G-mean [3], F-measure [2], or AUC [16]. However, all of them are designed especially for the binary class problems. Standard metrics such as classification rate (accuracy rate) is an unreasonable measure in the multi-class imbalance classification problem, as it does not differentiate the classification rates from different classes. For example, in the data set of thyroid, a classifier can achieve a high accuracy rate of 92.5%, if it recognizes all the instances as class 3. There are some proposals for measures displaying a balanced performance on multiple classes, like multi-class AUC [32]. For this study we have decided to use the average accuracy metric.

The average accuracy gives the same weight to each class. It achieves the accuracy rate of each class independently, and then the final result is obtained by the average value. The average accuracy is computed as follows:

$$\text{AveAcc} = \frac{1}{m} \sum_{i=1}^m \text{TRP}_i, \quad (4)$$

where m is the number of classes and TRP_i stands for the True Positive Rate of the i -th class.

4.4. Statistical analysis

Statistical tests are important for analyzing the experimental results to extract the findings. In this paper, the hypothesis testing techniques, which are recommended in [28], are used to provide statistical support for the analysis of the results obtained by

Table 2
Parameters setting for single and ensemble classifiers used in the study.

Methods	Parameters
Single classifiers	
CART	Prune = true Method = classification Impure nodes must have 10 or more examples to be split
BPNN	The number of nodes in hidden layer = 10 Transfer function of hidden layer = logsig Transfer function of output layer = logsig Train epochs = 100
SVM	$C = 1.0$ Tolerance parameter = 0.001 $\gamma = 0.0046$ Kernel type = Gaussian radial basis function Optimization method = SMO
Ensemble classifiers	
UBA	The number of bags = 40 The number of resampling majority class = the number of minority class
SBA	The number of bags = 40 The number of nearest neighbors in SMOTE = 5
RBO	The number of iterations in AdaBoost = 40 The number of resampling majority class = the number of minority class
SBO	The number of iterations in AdaBoost = 40 The number of nearest neighbors in SMOTE = 5
SMB	The number of iterations in AdaBoost = 40 The number of nearest neighbors in SMOTE = 5
Easy	The number of subsets is 4 The number of iteration in each AdaBoost ensemble is 10
Algorithms for multi-class imbalance	
OVA-NBSVM	$C = 2.0$ $\sigma \in [1, 2, 3, 4, 5, 10, 20, 30, 40, 50]$ - best selected for each dataset Optimization method = modified SMO classifier combination = OVA
AdaBoost.NC	The number of iterations = 51 $\alpha = 2$ base classifier = C4.5

the experiment. Specially, for the pairwise comparisons, Wilcoxon signed-rank test [60] is adopted as a non-parametric statistical procedure to perform pairwise comparisons between ensemble learning and resampling approaches in OVO scheme and comparisons of the representation ensemble learning with OVO scheme and the state-of-the-art methods. Then, for the multiple comparisons, average aligned-rankings [28] of each method are adopted to compare the behavior of each method with respect to the others. Moreover, Friedman aligned-ranks test is used to check whether the best method (the control one) is significant better than others.

5. Experimental study

In this section we develop a thorough empirical analysis in addressing the multi-class imbalance classification problems. We want to verify the effectiveness of the combination of ensemble learning and OVO scheme for the multi-class imbalance classification compared with classical approaches. Therefore, we develop the pairwise comparative study on analysis of ensemble learning approaches versus resampling techniques in the OVO scheme with different base classifiers, including CART, BPNN and SVM. Additionally, to show that using ensembles as base classifiers in OVO can efficiently empower learning from multi-class imbalanced data we present comparison with two state-of-the-art algorithms dedicated to this problem.

5.1. Evaluating ensemble approaches with CART as base classifier

The study for the CART decision tree is shown in Table 3, where we present all the results of average accuracy.

According to the best result stressed through bold-face, we can clearly find that the methods considering ensemble learning in the

OVO scheme always receive better results in each data set. Observing the average performance, SBO obtains the best performance, followed by SMB and SBA.

The statistical study based on Wilcoxon test for CART is developed in Table 4.

Results of Wilcoxon test show that for CART classifier ensemble systems can significantly outperform resampling-based strategies with single classifier. However, there is no single ensemble strategy that is statistically significantly better than all of three resampling approaches tested at the same time. For example SBO method achieves excellent p -values in comparison with RUS and SMOTE, but achieves p -value slightly above the significance level for comparison with ROS. Similar situation can be observed for all of highlighted best-performing ensemble techniques. At the same time one must notice that the OVO-based ensemble learning does not damage the accuracies when compared to OVO single model learning with resampling. Therefore, we can summarize that ensemble learning with OVO decomposition is a preferred choice over OVO resampling techniques, never leading to reduced classification rates and often offering significant improvement for the multi-class imbalance classification problems with CART classifier.

5.2. Evaluating ensemble approaches with BPNN as base classifier

The complete results for the BPNN version are shown in Table 5.

In this case the trend is quite similar with the segment of CART, since most of the best results obtained in each data set are acquired by the ensemble learning approaches. Additionally, for the average results, SMB receives the best performance, followed closely by SBO. This is consistent with those obtained by CART. Therefore, we must highlight that the methodology of ensemble

Table 3

Average accuracy [%] results for resampling and ensemble learning methods with CART as the base classifier. The best result is highlighted in bold.

Data sets	Resampling techniques			Ensemble learning					
	RUS	ROS	SMOTE	UBA	SBA	RBO	SBO	SMB	Easy
Aut	10.00	83.93	78.43	10.00	79.97	67.38	81.60	69.85	51.63
Bal	54.49	58.24	56.43	64.90	56.51	71.40	58.35	66.27	73.13
Car	93.87	94.64	91.93	95.72	93.87	96.53	97.34	97.56	97.57
Cle	31.92	30.19	28.82	31.92	29.22	32.98	31.65	33.18	31.89
Con	48.54	48.11	47.96	51.97	47.96	48.31	47.85	48.96	50.79
Der	94.36	95.15	95.37	95.45	95.37	94.49	94.94	94.65	94.41
Fla	63.28	62.52	61.20	63.46	61.11	62.65	60.58	62.60	63.03
Gla	64.08	66.66	67.90	71.38	69.39	69.68	72.17	72.59	71.78
Hay	86.07	85.86	85.76	85.86	85.76	85.54	85.15	85.24	86.15
Led	69.77	69.95	70.13	69.69	70.13	70.20	70.48	70.78	70.83
Lym	11.00	79.05	69.98	11.00	69.98	36.10	76.10	62.51	19.83
New	92.80	89.70	90.37	93.48	91.75	93.24	91.72	93.69	94.43
Pag	91.98	81.19	84.77	94.24	84.31	94.09	85.33	93.63	94.68
Sat	83.82	84.25	84.23	87.97	86.36	90.02	89.41	90.20	89.83
Shu	78.23	85.85	85.11	79.69	85.10	37.39	85.94	41.89	25.98
Spl	92.74	92.68	92.89	94.50	93.33	95.51	95.65	95.64	95.64
Thy	96.24	93.77	93.77	98.14	93.77	96.42	92.95	98.61	98.16
Win	91.77	92.60	92.29	94.18	92.29	97.32	91.34	96.60	95.63
Wqr	33.18	32.60	34.61	37.80	36.21	39.27	39.15	38.86	39.12
Zoo	48.01	85.67	72.15	49.23	72.15	48.08	87.48	62.64	23.67
Avg.	66.81	75.63	74.21	69.03	74.73	71.33	76.76	73.80	68.41

Table 4

Wilcoxon tests for comparison of ensemble learning and resampling techniques in the OVO scheme with CART as the base classifier. R^+ corresponds to the sum of the ranks for ensemble learning and R^- for the resampling techniques.

Comparison	R^+	R^-	p -value	Comparison	R^+	R^-	p -value
UBA vs. RUS	197.00	13.00	0.000629	SBO vs. RUS	157.00	53.00	0.052222
UBA vs. ROS	135.50	74.50	0.295424	SBO vs. ROS	143.00	67.00	0.156004
UBA vs. SMOTE	135.00	75.00	0.262722	SBO vs. SMOTE	179.00	31.00	0.005734
SBA vs. RUS	144.00	66.00	0.145400	SMB vs. RUS	185.00	25.00	0.002821
SBA vs. ROS	91.50	118.50	0.629162	SMB vs. ROS	131.00	79.00	0.331723
SBA vs. SMOTE	158.00	52.00	0.022909	SMB vs. SMOTE	137.00	73.00	0.232226
RBO vs. RUS	172.00	38.00	0.012374	Easy vs. RUS	167.00	43.00	0.020633
RBO vs. ROS	127.00	83.00	0.411465	Easy vs. ROS	133.00	77.00	0.295878
RBO vs. SMOTE	131.00	79.00	0.331723	Easy vs. SMOTE	133.00	77.00	0.295878

Table 5

Average accuracy [%] results for resampling and ensemble learning methods with BPNN as the base classifier. The best result is highlighted in bold.

Data sets	Resampling techniques			Ensemble learning					
	RUS	ROS	SMOTE	UBA	SBA	RBO	SBO	SMB	Easy
Aut	47.12	64.99	56.34	61.40	70.44	70.83	77.67	71.91	66.88
Bal	70.81	89.11	85.93	88.18	91.07	85.80	94.12	86.08	88.28
Car	61.78	72.10	66.98	77.17	80.41	90.27	97.98	95.93	91.79
Cle	33.03	35.18	32.18	32.62	30.07	32.68	30.70	30.31	31.16
Con	45.97	47.12	38.27	50.72	36.71	51.90	51.15	50.70	51.96
Der	91.60	93.81	92.49	96.87	96.81	96.41	96.94	96.90	96.79
Fla	56.96	60.07	54.11	65.24	59.61	63.38	62.28	63.75	65.05
Gla	48.91	52.67	51.39	63.21	60.02	71.09	63.92	70.91	70.02
Hay	60.17	61.84	53.36	71.06	66.53	77.78	82.49	78.38	78.71
Led	68.48	67.28	67.25	73.64	73.04	71.54	72.15	70.81	72.02
Lym	70.41	85.53	78.14	86.79	88.83	83.22	87.86	84.08	86.17
New	90.04	94.89	92.80	97.70	96.52	97.07	96.22	97.19	97.93
Pag	78.57	82.26	83.52	88.52	89.49	91.50	84.29	91.08	92.08
Sat	86.24	86.58	85.47	87.38	86.99	89.30	88.30	89.71	89.55
Shu	68.10	84.87	89.46	80.71	93.07	85.79	93.56	91.40	87.88
Spl	85.76	86.70	83.67	90.34	90.32	89.73	91.32	90.55	89.78
Thy	74.98	80.84	75.70	86.45	70.54	95.80	78.36	96.19	96.81
Win	94.08	92.09	95.50	98.70	98.27	97.94	98.00	97.87	98.57
Wqr	33.05	34.80	35.36	39.25	36.93	39.36	31.40	40.42	41.96
Zoo	73.04	81.19	79.61	90.24	90.10	86.60	89.52	88.86	89.39
Avg.	66.96	72.70	69.88	76.31	75.29	78.40	78.41	79.15	79.14

Table 6

Wilcoxon tests for comparison of ensemble learning and resampling techniques in the OVO scheme with BPNN as the base classifier. R^+ corresponds to the sum of the ranks for ensemble learning and R^- for the resampling techniques.

Comparison	R^+	R^-	p -value	Comparison	R^+	R^-	p -value
UBA vs. RUS	209	1	0.000103	SBO vs. RUS	206	4	0.000163
UBA vs. ROS	186	24	0.002495	SBO vs. ROS	186	24	0.002495
UBA vs. SMOTE	197	13	0.000593	SBO vs. SMOTE	201	9	0.000338
SBA vs. RUS	189	21	0.001713	SMB vs. RUS	208	2	0.00012
SBA vs. ROS	159	51	0.043804	SMB vs. ROS	196	14	0.000681
SBA vs. SMOTE	194	16	0.000892	SMB vs. SMOTE	208	2	0.00012
RBO vs. RUS	209	1	0.000103	Easy vs. RUS	209	1	0.000103
RBO vs. ROS	194	16	0.000892	Easy vs. ROS	199	11	0.000449
RBO vs. SMOTE	205	5	0.000189	Easy vs. SMOTE	207	3	0.00014

Table 7

Average accuracy [%] results for resampling and ensemble learning methods with SVM as the base classifier. The best result is highlighted in bold.

Data sets	Resampling techniques			Ensemble learning					
	RUS	ROS	SMOTE	UBA	SBA	RBO	SBO	SMB	Easy
Aut	47.86	56.74	58.01	52.30	57.63	60.71	67.34	67.51	60.41
Bal	61.78	57.43	73.70	64.12	73.66	78.77	64.60	80.59	79.64
Car	79.92	73.71	86.23	80.41	86.23	91.6	89.18	91.65	91.20
Cle	30.78	24.20	33.23	33.46	33.99	32.07	31.61	32.13	34.53
Con	50.37	46.32	48.40	50.76	48.40	50.12	41.61	50.35	50.85
Der	97.46	95.49	96.21	97.66	96.21	97.49	96.53	97.72	97.34
Fla	62.16	56.43	62.45	64.53	62.72	64.11	58.33	63.95	65.72
Gla	47.83	53.14	50.81	49.53	51.86	68.67	60.81	68.37	66.18
Hay	53.24	53.25	53.66	53.56	52.89	64.24	62.38	64.65	57.64
Led	70.95	67.93	70.11	73.52	70.11	71.78	72.88	72.10	72.31
Lym	83.12	89.26	85.16	85.94	88.50	83.35	87.80	82.96	87.35
New	85.25	90.00	92.20	87.30	91.91	96.13	95.66	95.43	97.70
Pag	71.30	76.84	79.63	73.15	80.46	89.87	83.84	90.95	90.74
Sat	86.24	71.59	86.81	86.32	86.86	84.94	84.32	85.30	86.21
Shu	68.27	94.15	94.73	74.64	94.70	77.97	99.33	88.72	85.46
Spl	95.96	95.43	96.30	96.16	96.43	95.94	95.86	96.07	96.20
Thy	68.27	79.49	82.61	70.48	82.74	91.11	78.74	92.46	93.67
Win	96.37	96.18	96.57	97.07	96.57	97.25	98.05	97.88	97.62
Wqr	37.20	35.53	38.64	38.57	38.83	40.63	24.67	39.41	38.92
Zoo	87.16	88.47	90.29	88.77	90.87	90.14	91.68	88.38	90.07
Avg.	69.07	70.08	73.79	70.91	74.08	76.34	74.26	77.33	76.99

learning methods used to train the binary classifier in the OVO scheme achieve quite high quality of the average performance.

The Wilcoxon tests are carried out in Table 6, when BPNN is used as the base classifier.

In this case, all the null hypotheses of equivalence are rejected for the ensemble learning methods, since the largest p -value is equal to 0.043804, which is lower than our α -value (0.05). This is much better than in the case of CART classifiers, as here we observe a globally statistically significant improvement regardless of the ensemble method used. This is a highly interesting observation, as commonly it was assumed that decision trees as weak learners should benefit the most from multiple classifier systems. However, one can see that combination of BPNNs works much better than combination of CARTs. Therefore, we can conclude that the ensemble learning approaches with OVO scheme for the multi-class imbalance classification problems outperform the OVO resampling techniques for BPNN classifier.

5.3. Evaluating ensemble approaches with SVM as base classifier

Finally, we show the complete results for the version of SVM in Table 7.

The average results obtained in this version are somewhat lower than in the case of BPNN, but it is consistent that the ensemble learning approaches improve the performance for the multi-class imbalance classification problems with the OVO scheme, comparing with the original resampling techniques. Additionally,

once again SMB is considered as the best average performance, followed by Easy and RBO respectively.

Observing the statistical analysis shown in Table 8, the behavior of the ensemble learning techniques in the OVO scheme is generally better than those obtained by the resampling approaches.

According to the statistical results of Wilcoxon test, we can observe that RBO, SMB and Easy clearly outperform all of the conventional resampling techniques. In addition, RUS is defeated by any ensemble learning approach. For ROS only UBA method does not deliver significant improvement. With regard to SMOTE the p -values observed for comparing with RBO, SMB and Easy are lower than our α -value (0.05). Therefore, in the scenario of SVM we also can conclude that ensemble learning approaches outperform resampling techniques in the OVO scheme for the multi-class imbalance classification problems.

5.4. Comparison of different used ensemble learning techniques

The goal of this study is to explore the validity of the ensemble learning approaches with OVO decomposition in the application of multi-class imbalance classification problems, since these techniques combine the single classifiers to be an ensemble, which is useful for addressing the binary-class imbalance problems. The facts found in this study support the goodness of the methodology of ensemble learning methods based on resampling techniques for improving the performance in dealing with the multi-class imbalance classification problems.

Table 8

Wilcoxon tests for comparison of ensemble learning and resampling techniques in the OVO scheme with SVM as the base classifier. R^+ corresponds to the sum of the ranks for ensemble learning and R^- for the resampling techniques.

Comparison	R^+	R^-	p -value	Comparison	R^+	R^-	p -value
UBA vs. RUS	210	0	0.000089	SBO vs. RUS	166	44	0.022769
UBA vs. ROS	129	81	0.370261	SBO vs. ROS	178	32	0.006425
UBA vs. SMOTE	62	148	0.108427	SBO vs. SMOTE	122	88	0.525653
SBA vs. RUS	186	24	0.002495	SMB vs. RUS	201	9	0.000338
SBA vs. ROS	196	14	0.000681	SMB vs. ROS	191	19	0.001325
SBA vs. SMOTE	145.5	64.5	0.111769	SMB vs. SMOTE	166	44	0.022769
RBO vs. RUS	197	13	0.000593	Easy vs. RUS	207	3	0.00014
RBO vs. ROS	183	27	0.003592	Easy vs. ROS	193	17	0.001019
RBO vs. SMOTE	165	45	0.025094	Easy vs. SMOTE	186	24	0.002495

Table 9

Friedman aligned-rank tests comparing the different ensemble learning methods with OVO scheme.

Ensemble learning	CART	BPNN	SVM
UBA	66.10(0.151106)	71.10(0.168688)	82.25(0.002173)
SBA	75.15(0.018658)	76.45(0.058981)	73.60(0.025194)
RBO	61.60(0.285839)	58.35(1.000000)	51.50(0.939695)
SBO	61.25(0.285839)	58.50(1.000000)	67.35(0.091477)
SMB	43.25	49.85(1.000000)	43.55
Easy	55.65(0.285839)	48.75	44.75(0.939695)

However, we also need to compare the different examined ensemble learning techniques among themselves in order to find the statistically best method for each base classifier. Using only averaged accuracy for such a selection can be misleading as we do not take into account the ranks among these methods for each dataset. Therefore, we need to conduct a statistical test over multiple datasets to choose the best combined classifier.

For this purpose we use the Friedman aligned-rank test in order to detect the highest-ranked methods. The results from this test are depicted in Table 9.

When analyzing the aligned ranks for CART classifier we can clearly see that SMB method achieves the best results. Its rank is clearly lower than other methods. This is especially interesting when considering the fact of SBO scoring the best average accuracy results. But according to statistical analysis SMB method displays the most efficient and stable performance over all of datasets used.

For BPNN classifier we can see that the lowest ranks were obtained by Easy approach. However, differences between it and SMB are practically insignificant (48.75 vs. 49.85) thus allowing us to conclude that both of these methods display excellent performance for neural networks in mining multi-class imbalanced data.

Similar situation can be observed for SVM classifier. Here SMB method obtains the lowest ranks but differences between it and Easy method are negligible (43.55 vs. 44.75), showing that both of these methods are highly suitable for working with SVMs.

5.5. Comparison with ad-hoc approaches for multi-class imbalanced data

In order to put the obtained results into context we cannot refer only to OVO-based resampling techniques. There is a number of solutions already proposed in the literature for addressing skewed distributions in multiple classes. We decided to select two ensemble-based techniques from them as reference methods for our proposal. We have selected popular AdaBoost.NC [57], which is considered as one of the best approaches for handling multi-class imbalanced data. Additionally, we present results for recently

introduced OVA-NBSVM [14] that combines binary modified SVMs using one-vs-all technique.

For each base classifier examined we have selected a single best-performing ensemble technique according to the Friedman aligned-rank tests discussed in Section 5.4. This means that for comparison SMB for CART/SVM and Easy for BPNN are being used.

Accuracy results for examined benchmarks and methods are depicted in Table 10.

We will focus our discussion on comparison with reference methods, as accuracies of ensemble techniques for different base classifiers were discussed in previous sections.

OVA-NBSVM returned surprisingly the worst results from all of the examined methods. Only for a single dataset (Zoo) it was able to outperform all other approaches. This proves the lack of usefulness of OVA decomposition for multi-class problems with skewed distributions. OVA for some of class combinations additionally boosts the disproportion between classes thus making learning from small-sample classes very difficult. This can be observed for cases with high imbalance ratios and small classes like Pag and Wqr datasets. Here some of classes are very small when compared to others, which leads to a extremely high binary imbalance ratio when these classes are used as a positive and aggregation of remaining ones as a negative. For these cases OVA-NBSVM drops highly in comparison to all other discussed solutions.

AdaBoost.NC is a more demanding reference method, achieving best results on four datasets. However, on 16 remaining benchmarks one of the proposed ensemble techniques outperforms this boosting scheme. It allows us to conclude that empowering OVO decomposition with ensemble learning can be a highly efficient solution for the discussed problem, being able to deliver generally improved performance over most popular solutions from the literature.

To gain an additional insight into the performance of examined methods we have conducted a Wilcoxon test, outcomes of which are presented in Table 11.

Obtained p -values show that proposed ensemble techniques with BPNN and SVM are significantly better than OVA-NBSVM. For comparison with AdaBoost.NC only BPNN-based ensemble offer significantly superior results. When analyzing obtained p -values and accuracies for remaining methods we can see that they offer small but visible improvement over reference algorithms. In conclusion it must be noted that the proposed combination of OVO and pairwise ensemble learning can outperform state-of-the-art methods for multi-class imbalance on a variety of datasets, thus being a worthwhile choice for such problems.

6. Lessons learned

In order to summarize this manuscript let us present three main points capturing the most important research findings.

Table 10

Complete average accuracy test results for the representative combinations (SMB-CART, Easy-BPNN and SMB-SVM) and the state-of-the-art approaches for mining multi-class imbalanced data (AdaBoost.NC and OVA-NBSVM). The best result is highlighted in bold.

Dataset	AdaBoost.NC	OVA-NBSVM	SMB-CART	Easy-BPNN	SMB-SVM
Aut	76.53	70.73	69.85	66.88	67.51
Bal	66.92	85.81	66.27	88.28	80.59
Car	85.08	54.83	97.56	91.79	91.65
Cle	30.33	31.11	33.18	31.16	32.13
Con	47.36	47.21	48.96	51.96	50.35
Der	95.00	96.45	94.65	96.79	97.72
Fla	60.57	54.45	62.60	65.05	63.95
Gla	69.89	61.96	72.59	70.02	68.37
Hay	85.39	76.42	85.24	78.71	64.65
Led	72.10	59.45	70.78	72.02	72.10
Lym	78.33	69.86	62.51	86.17	82.96
New	88.42	89.05	93.69	97.93	95.43
Pag	80.75	48.97	93.63	92.08	90.95
Sat	87.49	88.16	90.20	89.55	85.30
Shu	89.82	82.07	41.89	87.88	88.72
Spl	94.14	74.10	95.64	89.78	96.07
Thy	95.46	64.97	98.61	96.81	92.46
Win	93.13	97.58	96.60	98.57	97.88
Wqr	35.73	31.84	38.86	41.96	39.41
Zoo	86.67	90.62	62.64	89.39	88.38
Avg.	75.96	68.78	73.80	79.14	77.33

Table 11

Wilcoxon tests for comparison of the representative ensemble learning with OVO scheme and the state-of-the-art methods for multi-class imbalanced data. R^+ corresponds to the sum of the ranks for the representative ensemble learning and R^- for the state-of-the-art methods.

Comparison	R^+	R^-	p -value
SMB-CART vs. AdaBoost.NC	128	82	0.390533
Easy-BPNN vs. AdaBoost.NC	162	48	0.033340
SMB-SVM vs. AdaBoost.NC	150	60	0.092963
SMB-CART vs. OVA-NBSVM	146	64	0.125859
Easy-BPNN vs. OVA-NBSVM	198	12	0.000517
SMB-SVM vs. OVA-NBSVM	172	38	0.012374

- 1. The role of ensemble learning method.** Obtained results allow us to observe the existence of a trend among six examined ensemble learning techniques. Regardless of the used base classifier SMB and Easy methods delivered the best performance both in terms of averaged accuracy and ranking statistical tests. This makes them the most universal ensemble techniques that should always be examined in combination with OVO decomposition when imbalanced multi-class problems are being faced. Additionally, RBO and SBO techniques tend to deliver good results for some of the datasets, thus making them a reasonable second choice if time allows for a more compound experimental study.
- 2. The role of base classifier.** The choice of a base classifier had a significant impact on the observed classification accuracies. Surprisingly, used decision tree model (CART) returned least satisfactory improvement when used in ensemble setting. This is contrary to numerous statements in the literature, reporting excellent properties of tree models as weak classifiers for combination. This may be explained by high sensitivity of CART model to skewed distributions. SVM worked very well with most of the ensemble techniques, achieving improved classification rates in most of scenarios. BPNN was found to be the best working model in ensemble setting, achieving statistically superior results for any kind of used committee approach.
- 3. Comparison with resampling and ad-hoc solutions.** The proposed empowering of OVO decomposition with pairwise ensemble learning achieved highly competitive results when compared with standard OVO with resampling techniques applied.

Using compound learners for each pair of classes can lead to a better capturing of their local specifics, higher robustness to imbalance and improved final multi-class recognition rates. Additionally, proposed ensemble-based OVO is able to outperform state-of-the-art methods for multi-class imbalanced data, especially those using OVA solutions.

7. Conclusions and future works

In this paper we have proposed to improve the performance of binary decomposition used for multi-class imbalanced problems by applying ensemble learning paradigm to each sub-problem. This way we are able to exploit highly efficient combined classification approaches that were so far restricted to binary scenarios. Their proven efficacy in two-class imbalanced tasks motivated us to extend their area of applicability to much more challenging scenarios, where multiple majority and minority classes are present and the relationships between them are no longer obvious. As we wanted to capture pairwise relations between objects we focused one OVO decomposition, as it does not affect the distributions of positive and negative classes.

To check the flexibility of the proposed approach we have tested it with three different base classifiers: CART, BPNN and SVM. Experimental study, backed-up by a thorough statistical analysis indicate that it is possible to significantly boost the OVO approach performance for multi-class imbalanced data when enhancing it with ensemble classifiers. Regardless of the used committee approach in most cases we were able to outperform traditional OVO approach utilizing single classifiers with pre-processing algorithms, as well as state-of-the-art multi-class solutions for skewed data.

Obtained results allow us to formulate recommendations for selecting ensemble schemes. SMB technique is the best choice when considering CART and SVM as base classifiers. At the same time BPNN should be used with Easy approach. As result differences between several methods are not high, we point out the high effectiveness of Easy, SMB and RBO ensemble methods for empowering OVO techniques in multi-class imbalance scenarios.

We highlighted the effectiveness of synergy between decomposition strategy and ensemble learning in the multi-class imbalanced datasets. Above all, we must emphasize that our work provides the basis for addressing the multi-class imbalance

classification problems with a combination of decomposition and multiple classifier approaches.

In our study we only presented the VOTE aggregation strategy. Therefore, it is interesting to develop the analysis of the other aggregations in the OVO scheme and trained combiners like Error-Correcting Output Codes or Decision Templates. Additionally, we would like to extend our proposal to include recent findings in OVO decomposition that take into consideration the dynamic classifier selection [25] and competence-based weighting [26] to remove the non-competent classifiers from the pool.

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